Engagement Maximization

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Introduction

Motivation: online marketing platforms

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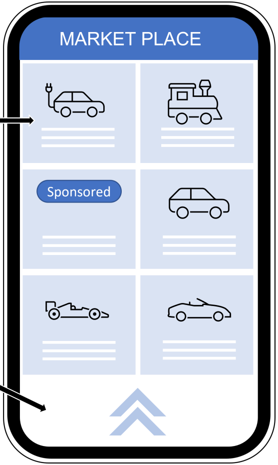
  - Instrumental / recreational value for user

- Browsing:
  - Controlled by user
  - Stops when value-added is low



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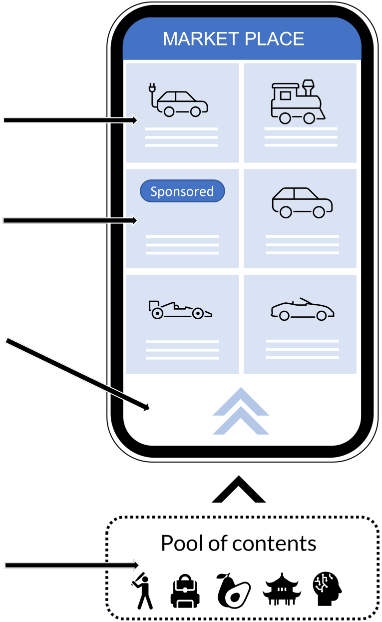
    - Maximize matching quality



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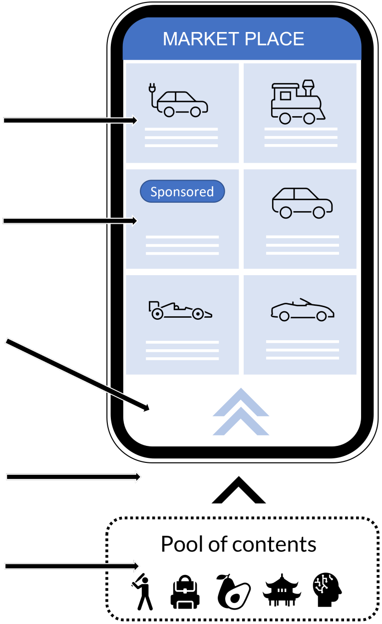

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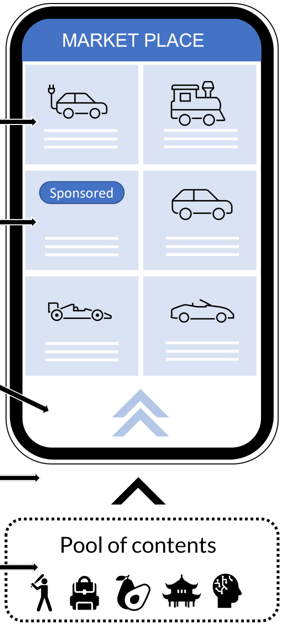
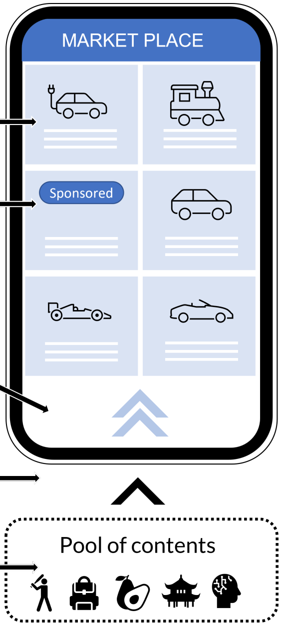
  - *Maximize matching quality* ← incentive misaligned



Introduction

What We Do

- Contents:
  - *Signal about unknown state*
- Ad impressions:
  - *"Attention" → Revenue*
- Browsing:
  - *Time is costly*
  - *Stop when continuation value is low*
- Information bottleneck:
  - *"Attention" constraint*
- Algorithm:
  - *Flexibly designs signal*
  - *Maximize revenue*



Introduction

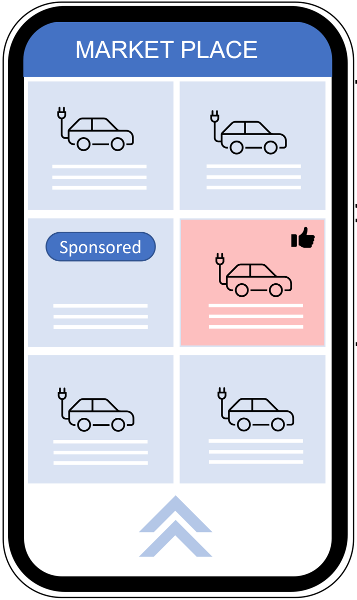
What We Find

Contents:

- *Implemented by stationary Poisson signals ---Deep & Narrow*
- *Deeper & narrower than user's first best*

Browsing:

- *Zero consumer surplus*
- *Longer stopping time than user's first best*



Suspensive Signals:

- *Induces browsing*
- *Makes user more "uncertain" than prior*

Decisive Signals:

- *Induces stopping*
- *Reduce to static RI model*

Introduction

Related Literatures

- User-optimal benchmark
  - Dynamic RI: Zhong 2022, Hébert and Woodford 2023.
  - Equivalent static RI model: Matejka and McKay 2015, Caplin and Dean 2015.
- Information bottleneck
  - Uniformly posterior separable: Caplin, Dean, and Leahy 2022, Bloedel and Zhong 2020
- Unconstrained version
  - Dynamic info. design: Orlov, Skrzypacz, and Zryumov 2020, Ely and Szydlowski 2020

Model

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  - *Unknown state* $x \in X$, *prior* q_0
  - *Belief process* $\langle q_t \rangle$

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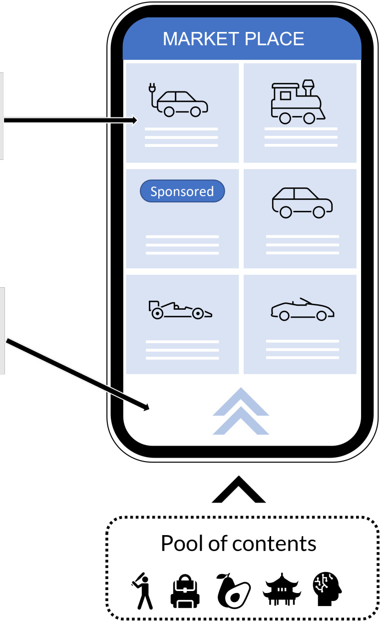
  - *Stopping utility* $u(q)$
  - $\tau \in \arg\max_\tau E^P[u(q_\tau) - \kappa\tau)]$



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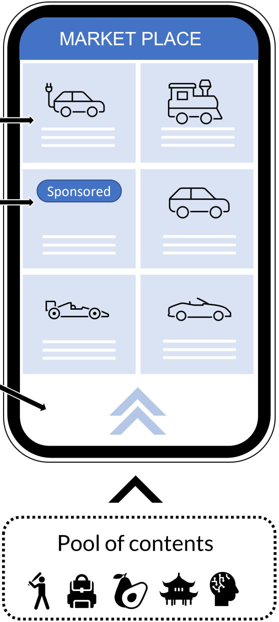

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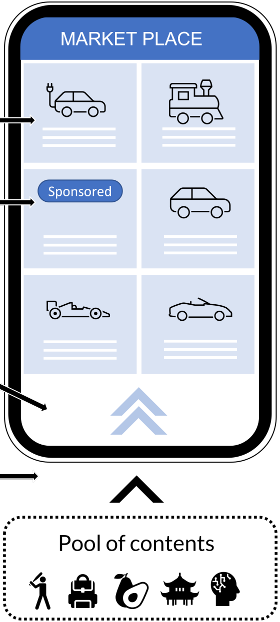

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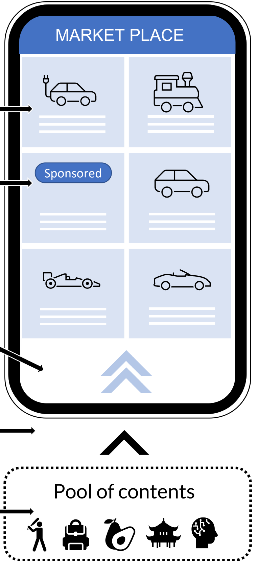
- Algorithm:
  - $\sup_{\langle q_t \rangle, \tau} E^P[G(q_\tau) - G(q_0)]$
    - *s.t. IC & info. constraints*



Model

Simplified Example

- Contents:
  - Unknown state $x = L$ or R, uniform prior.
  - Signal $s = s_L$ or s_R with arrival rate:

	L	R
s_L	$\alpha \cdot p$	$\alpha \cdot (1 - p)$
s_R	$\alpha \cdot (1 - p)$	$\alpha \cdot p$

    - Parameters: α–broadness, p–depth

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- Information bottleneck:
  - Mutual info. rate: $\alpha \cdot \underbrace{\left(H^S(0.5) - H^S(p)\right)}_{\text{Increase fn. of } |p-0.5|} \leq 1.$
  - Broadness-depth tradeoff.

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- User's decision problem: action $a = l$ or r,
  - Utility: $1_{a=x} - 1_{a \neq x} - 2t$

Model

Simplified Example

- Primitives
  - Contents:
    - State $x = L$ or R, uniform prior.
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- User's optimum:
  - $\sup_{\alpha,p} (2p - 1) - \frac{2}{\alpha}$ s.t. $\alpha \cdot \left(H^S(0.5) - H^S(p)\right) \leq 1$
  - $p = \frac{e}{1+e}$ and $\alpha = \left(H^S(0.5) - H^S(p)\right)^{-1}$.

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  - $p = \frac{e}{1+e}$ and $\alpha = \left(H^S(0.5) - H^S(p)\right)^{-1}$.
- A better (near optimal) strategy for the platform:
  - $\widetilde{p} = \frac{4e}{1+4e}$ and $\widetilde{a} = \left(H^S(0.5) - H^S(\widetilde{p})\right)^{-1}$

Model

Simplified Example

- User's optimum (assume info. bottleneck binding):
  - $\sup_{p} (2p - 1) - 2\left(H^S(0.5) - H^S(p)\right)$

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- Platform's optimum:
  - $\sup_p \left(H^S(0.5) - H^S(p)\right)$
    s.t. $2p - 1 - 2\left(H^S(0.5) - H^S(p)\right) \geq 0.$

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$\implies$ $\inf_\lambda \sup_p \lambda(2p-1) - (2\lambda - 1)\left(H^S(0.5) - H^S(p)\right).$

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  - Stationarity of signal guarantees IC.

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1. Analogous to static problems (extreme beliefs)
2. Zero surplus to user (welfare minimization)
3. Belief remains at prior until stopping (suspensive / decisive signals)

Solution

The General Problem

- Platform's problem:

$$\sup_{\langle q_t \rangle, \tau} \mathbb{E}\left[G(q_\tau) - G(q_0)\right] \tag{1}$$

$$\text{s.t.} \begin{cases} \tau \in \arg\max_{\tau'} \mathbb{E}\left[u(q_\tau) - \kappa\tau\right] \\ \frac{\mathrm{d}\mathbb{E}\left[H(q_t)\middle|\mathcal{F}_t\right]}{\mathrm{d}t} \leq \chi \\ \langle q_t \rangle \text{ is a martingale} \end{cases}$$

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Assumption 1

$u(q_0) - \frac{\kappa}{\chi} H(q_0) > \sum_x q_0(x) \left(u(e_x) - \frac{\kappa}{\chi} H(e_x) \right)$, where e_x is the degenerate belief on x.

- "Learning everything is worse than learning nothing"— avoids a corner case.

Solution

Information Costs and Benefits

- H governs the cost of processing information for the user
- G governs the benefit to the platform of sending the information
- Two leading cases:
  1. $G{=}H$: platform doesn't care what user learns, only that engagement maximized
     - Interpretation: ads are mixed with content in non-separable way
     - or platform is paid based on time spent using platform
  2. $X = X_1 \times X_2$, G depends on X_2, $\hat{u}$ depends on X_1
     - X_2 are the ads, platform wants them to get attention
     - X_1 is the content the user wants

Solution

The Relaxed Problem

- Platform's problem:

$$\sup_{\langle q_t\rangle,\tau} \mathbb{E}\left[G(q_\tau) - G(q_0)\right] \tag{(1)}$$

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- Observation: $\forall$ admissible $(\langle q_t\rangle, \tau)$, let $\pi \sim q_\tau$
  - $\underbrace{\mathbb{E}_\pi\left[u(q) - u(q_0)\right] \ge \kappa\mathbb{E}_\pi[\tau]}_{\text{ex ante IR}} \overbrace{\ge \frac{\kappa}{\chi}\mathbb{E}_\pi\left[H(q) - H(q_0)\right]}^{\text{aggregate info. constraint}}.$

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- Relaxed problem:

$$\sup_{\pi,\mathbb{E}_\pi[q]=q_0} \mathbb{E}_\pi[G(q) - G(q_0)] \tag{2}$$

$$\text{s.t. } \mathbb{E}_\pi[u(q) - u(q_0)] \geq \frac{\kappa}{\chi}\mathbb{E}_\pi[H(q) - H(q_0)].$$

Solution

The Relaxed Problem

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Proposition 1

Equation (2) has a solution π^* with finite support. When equation (2)> 0, there exists $\lambda > \frac{\chi}{\kappa}$ s.t. all such π^* maximizes

$$\mathbb{E}_\pi\left[u(q) - \frac{\kappa}{\chi}(H(q) - H(q_0)) + \frac{1}{\lambda}(G(q) - G(q_0))\right].$$

- π^* solved via concavifying $u - \left(\frac{\kappa}{\chi} - \frac{1}{\lambda}\right) H$.

Solution

Implementation

- Relaxed problem:

$$\sup_{\pi, \mathbb{E}_\pi[q]=q_0} \mathbb{E}_\pi[H(q) - H(q_0)] \tag{(2)}$$
$$\text{s.t. } \mathbb{E}_\pi[u(q) - u(q_0)] \geq \frac{\kappa}{\chi}\mathbb{E}_\pi[H(q) - H(q_0)].$$

Proposition 2

$\forall$ non-degenerate π that is admissible in equation (2), let $\alpha = \frac{\chi}{\mathbb{E}_\pi[H(q)-H(q_0)]}$, $Q \sim \pi$ and

$$\begin{cases} q_t = q_t + 1_{N_\alpha(t)\geq 1} \cdot (Q - q_0) \\ \tau = \min\{t | N_\alpha(t) \geq 1\}. \end{cases}$$

$(\langle q_t \rangle, \tau)$ is admissible in equation (1).

 – We say $\langle q_t \rangle$ is an α-dilution of π (Pomatto, Strack, and Tamuz 2023).

Solution

Main theorem

- Combining Propositions 1 and 2 immediately implies

Theorem 1

$\forall q_0$, there are two cases:

1. π^* solving equation (2) is degenerate. The user immediately stops and any feasible policy is optimal.
2. π^* solving equation (2) is non-degenerate. The α-dilution of π^* implements the platform's optimum.

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2. π^* solving equation (2) is non-degenerate. The α-dilution of π^* implements the platform's optimum.

- Comments:
  - User's ex ante IR binding — *full surplus extraction*
  - Stationary content flow — *commitment is unnecessary*
  - Poisson type signals — *narrow & deep content*
  - So far, only sufficiency for $\langle q_t \rangle$

Characterization of optimal policies

Extreme Beliefs

- User's optimal content flow:
  - $(\langle q_t^U \rangle, \tau^U) \in \arg\max_{\langle q_t \rangle, \tau} \mathbb{E}\left[u(q_\tau) - \kappa\tau\right]$ s.t. $\mathrm{d}\mathbb{E}\left[H(q_t)|\mathcal{F}_t\right] \leq \chi \mathrm{d}t$.

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  - $\iff \; q_\tau \sim \pi^U \in \arg\max_{\pi} \mathbb{E}_\pi \left[u(q) - \frac{\kappa}{\chi}(H(q) - H(q_0))\right]$.

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  - Let Q^U be the union of support of all user-optimal π^U.

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  - Let Q^U be the union of support of all user-optimal π^U.
- Let Q^P be the union of support of all platform-optimal π^*.
  - $\pi^* \in \arg\max \mathbb{E}_\pi \left[u(q) - \frac{\kappa}{\chi}(H(q) - H(q_0)) + \frac{1}{\lambda}(G(q) - G(q_0))\right]$.

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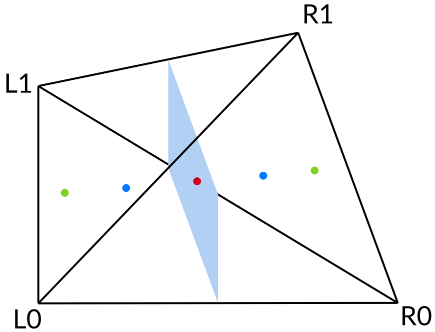
Proposition 3

$Q^P \bigcap \mathrm{Conv} Q^U = \emptyset$.

Leading Examples #1



- Prior belief
- Agent-optimal posteriors
- Principal-optimal posteriors

Figure: When $H = G$: ”Give em even more of what they want”

Leading Examples #2

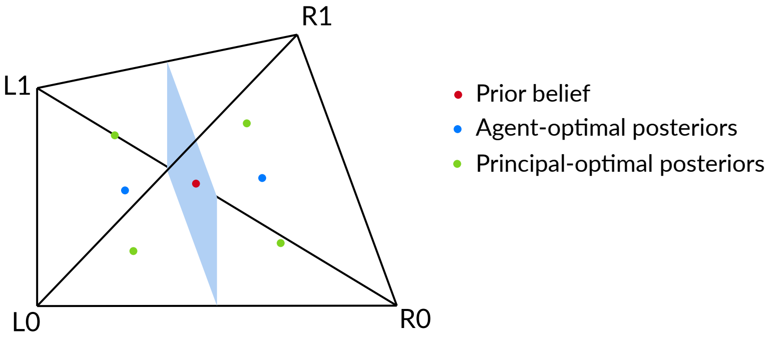


Figure: When $G \perp U$: "mix ads and content"

Characterization of optimal policies

Decisive and Suspensive Beliefs

- $V_R(q,\pi) = \max_{\pi' \in \mathscr{P}(\mathrm{Supp}(\pi)), \mathbb{E}_{\pi'}[q']=q} E_{\pi'}\left[u(q') - u(q) - \frac{\kappa}{\chi}(H(q') - H(q))\right].$
  - User's optimum when support constrained by π.
  - $V_R(q_0,\pi^*) = 0 \implies V_R(q,\pi^*)$ positive requires q *more uncertain* than prior.

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Definition 1

$q \in \mathscr{P}(X)$ is

1. Decisive: if $q \in Q^P$.
2. Suspensive: if not decisive & $\exists \pi^*$ s.t. $V_R(q,\pi^*) \geq 0$, denoted by Q^S.

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Proposition 4

$\forall$ solution $(\langle q_t \rangle, \tau)$ to equation (1),

- Continuation belief $\bigcup_{t\in[0,\tau)} \mathrm{Supp}(q_t) \subset Q^S$;
- Stopping belief $\mathrm{Supp}(q_\tau) \subset Q^P$.

Characterization of optimal policies

Simplified example

- Primitives
  - Contents:
    - State $x = L$ or R, prior q_0.
  - Information bottleneck: $\mathbb{E}\left[\mathrm{d}H^S(q_t)\middle|\mathcal{F}_t\right] \leq \mathrm{d}t$.
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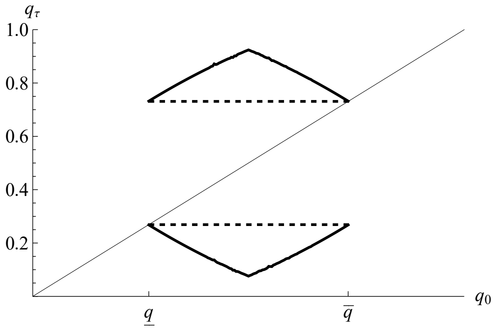

Figure: Q^U and Q^P as correspondences of q_0

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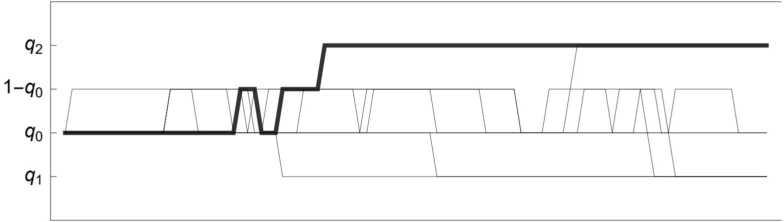
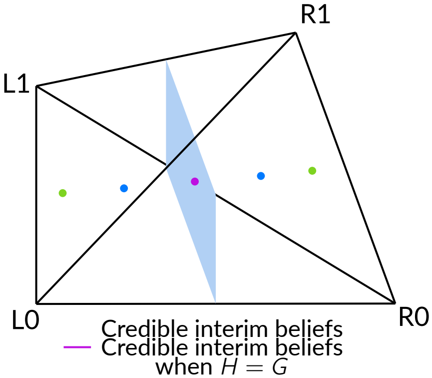


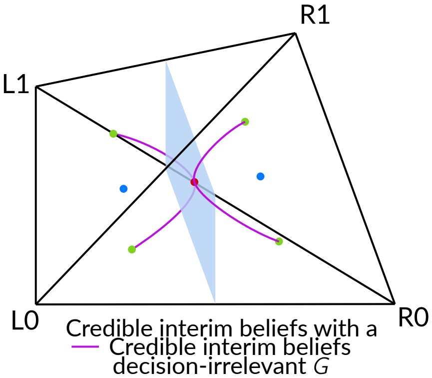
Figure: Sample paths of an optimal policy

Credibility

- Maintaining $V_R > 0$ (equivalently, $\hat{u}(q_t) > \hat{u}(q_0)$) requires commitment
- When the value of information for the agent rises, principal is tempted to reoptimize
- Result: only paths where $\hat{u}(q_t) = \hat{u}(q_0)$ are equilibria when principal can't commitment
- Signals are suspensive, but never strictly suspensive
- i.e. principal has same payoffs with and without commitment, but more limited set of policies

Leading Examples





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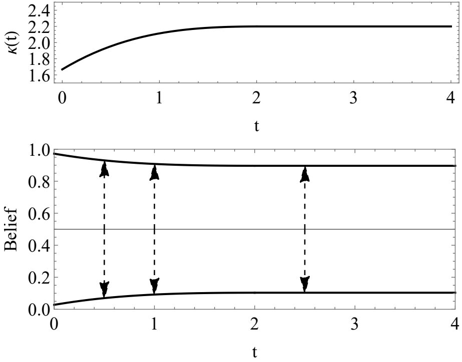

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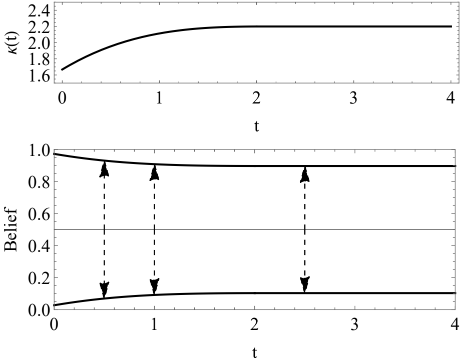


3. Bounded belief jump size:
   - Continuous path + $|X| = 2$: user's optimum

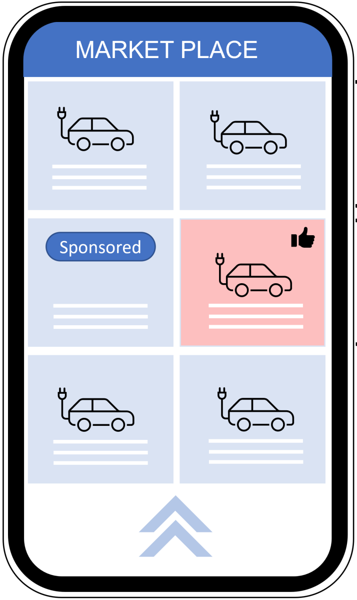
Conclusion

Contents:

- *Implemented by stationary Poisson signals ---Deep & Narrow*
- *Deeper & narrower than user's first best*

Browsing:

- *Zero consumer surplus*
- *Longer stopping time than user's first best*



Suspensive Signals:

- *Induces browsing*
- *Makes user more "uncertain" than prior*

Decisive Signals:

- *Induces stopping*
- *Reduce to static RI model*

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